

HEALTHCARE FACILITY & PATIENT MANAGEMENT SYSTEM

A Detailed Data Analysis Report

Diabetes Dataset — Exploratory & Descriptive Analysis

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Dataset: Diabetes Dataset (diabetic_data.csv) — Kaggle

Repository: <https://github.com/praneeth-tech/HF-PMS.git>

Report Information

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1. Executive Summary

This report presents a comprehensive data analysis of the Diabetes Dataset obtained from Kaggle, conducted as part of the Healthcare Facility & Patient Management System internship project (June–July 2026). The analysis was performed over the course of one week, progressing methodically from project environment configuration and dataset acquisition through to exploratory data analysis and key findings documentation.

The Diabetes Dataset comprises 101,766 hospital encounter records spanning 50 features, representing a rich cross-section of patient admission, treatment, and discharge information. The dataset includes 13 numerical and 37 categorical attributes, encompassing clinical measurements, medication records, and demographic variables pertinent to diabetic patient management.

The analysis was structured across six analytical phases: project environment setup, dataset loading and understanding, data quality assessment, exploratory data analysis (EDA), insight generation, and key findings documentation. Each phase was documented in a dedicated Jupyter notebook and synchronised with a version-controlled GitHub repository.

Key outcomes of the analysis include the identification of two critically incomplete features (`max_glu_serum` and `A1Cresult` with missing rates exceeding 83%), the discovery of clinically meaningful correlations between treatment complexity indicators, and the characterisation of pronounced skewness in patient visit frequency distributions. Ten structured key findings have been compiled to serve as the foundation for the final analysis report and future predictive modelling activities.

2. Introduction

2.1 Project Background

The Healthcare Facility & Patient Management System project was undertaken as part of an internship programme designed to provide practical experience in data analysis, machine learning preparatory work, and healthcare informatics. The project focuses on the application of data science methodologies to a real-world healthcare dataset with the objective of deriving actionable insights that may support clinical decision-making and patient outcome research.

The work was conducted entirely within a Python-based analytical environment, utilising Google Colab as the primary notebook platform, with all outputs systematically tracked and version-controlled via Git and GitHub. Daily progress was documented through structured reports, capturing source code, outputs, methodology, and analytical observations.

2.2 Dataset Description

The primary analytical resource for this project is the Diabetes Dataset, sourced from the Kaggle data repository. This dataset was compiled from clinical records of diabetic patients and contains information spanning multiple dimensions of the hospital encounter lifecycle.

The dataset file, `diabetic_data.csv`, is structured with 101,766 rows representing individual patient encounter records and 50 columns representing distinct clinical and administrative features. These features encompass patient identifiers, demographic attributes (age, gender, race), admission and discharge details, clinical measurements (laboratory procedure counts, diagnosis codes), medication records (including 24 specific drug-related columns), and readmission status.

2.3 Objectives of the Analysis

The analysis was guided by the following overarching objectives:

- To develop a thorough understanding of the structure, composition, and quality of the*

Diabetes Dataset.

- *To identify missing values, duplicate records, and feature-level data quality issues that may affect analytical validity.*
- *To explore the distributional characteristics and statistical properties of numerical and categorical features through visualisation and summary statistics.*
- *To investigate correlations and relationships between healthcare-related variables using formal correlation analysis.*
- *To extract meaningful healthcare insights from the data and compile a structured set of key findings to inform future modelling activities.*
- *To establish a well-documented, reproducible analytical workflow underpinned by version control best practices.*

2.4 Tools and Technologies

The following tools and technologies were employed throughout the analysis:

- **Python 3** — primary programming language for all analytical tasks.
- **Pandas** — data loading, manipulation, and summary statistics generation.
- **NumPy** — numerical computation and array operations.
- **Matplotlib** — static visualisation and plot generation.
- **Seaborn** — statistical visualisation including heatmaps, KDE plots, and pair plots.
- **Google Colab** — cloud-based Jupyter notebook environment.
- **Git / GitHub** — version control and remote repository management.
- **Overleaf** — LaTeX-based documentation and report preparation platform.

3. Project Environment Setup

3.1 Repository Structure

Upon project initiation on 01 June 2026, a structured GitHub repository named *HF-PMS* was established at <https://github.com/praneeth-tech/HF-PMS.git>. The repository was organised into five principal directories to ensure separation of concerns across different types of project artefacts:

- *data/* — contains the primary CSV dataset file (*diabetic_data.csv*).
- *notebooks/* — houses all analytical Jupyter notebooks, one per analytical phase.
- *reports/* — stores daily and weekly progress reports.
- *visualizations/* — contains generated plots and graphical outputs.
- *documentation/* — maintains project documentation and reference materials.

Each directory was initialised with a *.gitkeep* file to allow Git tracking of otherwise empty folders. The *README.md* file was created to provide a project overview, dataset source reference, tool listing, and authorship information.

3.2 Version Control Workflow

A consistent version control workflow was maintained throughout the project duration. Each analytical session concluded with the following Git operations:

1. *git add .* — staging all modified and newly created files.
2. *git commit -m "[descriptive message]"* — committing staged changes with a meaningful commit message.
3. *git push* — pushing committed changes to the remote GitHub repository.

This workflow ensured that all incremental analytical work was captured, documented, and recoverable. Commit messages were written to accurately reflect the nature of the work completed

in each session, providing a transparent audit trail of project development.

4. Dataset Understanding

4.1 Dataset Dimensions and Structure

The Diabetes Dataset was loaded into the analytical environment on 02 June 2026 using the Pandas `read_csv()` function. Initial inspection of the dataset revealed the following structural characteristics:

Table 4.1. Dataset Summary Statistics

Metric	Value
Total Rows (original dataset)	101,766
Total Columns / Features	50
Numerical Features	13
Categorical Features	37
Total Missing Values (original)	178,428
Duplicate Records	0
Missing Value — <i>max_glu_serum</i>	94,899 (94.68%)
Missing Value — <i>A1Cresult</i>	83,483 (83.29%)
Unique Patient Numbers	71,518
Memory Usage	38.8+ MB

The dataset spans 101,766 encounter records across 50 features, with a combined memory footprint of 38.8+ MB. Each row represents a unique hospital encounter identified by a distinct *encounter_id* value. The *patient_nbr* field yields 71,518 unique patient numbers, indicating that some patients contributed multiple encounter records to the dataset.

Initial structural inspection was performed using standard Pandas exploration functions including `df.head()`, `df.tail()`, `df.sample()`, `df.shape`, `df.info()`, `df.dtypes`, and `df.describe()`.

4.2 Feature Classification

A systematic feature classification was performed to distinguish between numerical and categorical attribute types. Thirteen features were identified as numerical (integer or float64 dtype), while the remaining 37 features were classified as categorical (object dtype).

The 13 **numerical features** include: *encounter_id*, *patient_nbr*, *admission_type_id*, *discharge_disposition_id*, *admission_source_id*, *time_in_hospital*, *num_lab_procedures*, *num_procedures*, *num_medications*, *number_outpatient*, *number_emergency*, *number_inpatient*, and *number_diagnoses*.

The 37 **categorical features** encompass demographic attributes (race, gender, age, weight), clinical identifiers (payer_code, medical_specialty), diagnosis codes (diag_1, diag_2, diag_3), laboratory result indicators (max_glu_serum, A1Cresult), 24 medication-related binary or ordinal columns, and the target variable readmitted.

4.3 Summary Statistics

Descriptive statistical analysis was conducted using `df.describe()` for numerical features. Key observations include:

- The mean time in hospital is approximately 4.40 days (SD = 2.99; max = 14 days).
- The average number of laboratory procedures per encounter is 43.11 (SD = 19.10; max = 132).
- *num_medications* has a mean of 15.30 and a maximum of 81, indicating wide variation in medication complexity.
- *number_emergency* has a mean of 0.12, confirming that emergency visits are rare but can reach a maximum of 42.
- The readmitted target variable contains three categories: NO, >30, and <30 days.

5. Data Quality Assessment

5.1 Missing Value Analysis

A comprehensive missing value assessment was conducted on 03 June 2026, employing `df.isnull().sum()` to identify columns containing null entries, followed by quantitative analysis of missing counts and percentage rates.

The `max_glu_serum` feature, representing maximum glucose serum measurement, contains 94,899 missing entries (94.68% missing rate). This indicates that glucose serum measurements were recorded for only approximately 5.32% of all encounters. Similarly, the `A1Cresult` feature has 83,483 missing values (83.29% missing rate), with valid entries present in only 17,018 records.

It is important to note that the post-preprocessing version of the dataset (used during EDA on Days 3–5, comprising 27,793 rows) exhibits 47,483 total missing values, suggesting significant filtering and transformation was applied between the raw loading phase and the analytical phase.

A missing values heatmap was generated using Seaborn to visually confirm that missing values are concentrated almost exclusively in `max_glu_serum` and `A1Cresult`.

5.2 Duplicate Record Detection

Duplicate record detection was performed using `df.duplicated().sum()`. The analysis confirmed **zero duplicate records**, validating the integrity of the encounter-level data collection process.

5.3 Feature Completeness Evaluation

A feature quality summary table was constructed consolidating data types, missing value counts, and unique value cardinalities for all 50 features. Key observations include:

- *encounter_id* contains 101,766 unique values, confirming its role as a unique encounter identifier.
- *diag_1*, *diag_2*, and *diag_3* contain 717, 749, and 790 unique values respectively, reflecting the breadth of clinical conditions represented.
- *medical_specialty* records 73 distinct values across patient admissions.
- Medication-related features generally contain 2–4 unique values (No, Down, Steady, Up).

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed on 04 June 2026, employing a suite of statistical and visual techniques to uncover distributional characteristics, detect anomalies, and investigate relationships among numerical features.

6.1 Distribution Analysis — Histograms

Histograms were generated for all 13 numerical features using `df.hist()` with a bin count of 20. Key observations:

- `encounter_id` and `patient_nbr` display approximately uniform distributions, consistent with their identifier roles.*
- `time_in_hospital` shows a right-skewed distribution with peak frequency in the 2–4 day range.*
- `num_lab_procedures` displays a near-normal distribution centred around 40–50 procedures.*
- `num_medications` follows a right-skewed distribution peaking in the 10–20 range.*
- `number_outpatient`, `number_emergency`, and `number_inpatient` exhibit extreme right-skew with values concentrated near zero.*
- `number_diagnoses` displays a multi-modal distribution with distinct peaks, particularly at values 7–9.*

6.2 Density Estimation — KDE Plots

Kernel Density Estimation (KDE) plots were generated for each numerical feature using Seaborn's `kdeplot()`. The KDE analysis reinforced histogram observations: `number_emergency` and `number_outpatient` showed sharp spike-like density peaks at zero, `num_medications` revealed a clean unimodal distribution, and `time_in_hospital` presented a characteristic right-skewed wave pattern reflecting its discrete integer nature.

6.3 Outlier Detection — Boxplot Analysis

Boxplot visualisations were constructed for all numerical features using `sns.boxplot()`. Key findings:

- `num_lab_procedures` exhibits a dense cluster of outlier points above the upper whisker (range: 90–132 procedures).
- `num_medications` shows outlier markers ranging from approximately 32 to 81, reflecting highly complex pharmacological cases.
- `number_outpatient`, `number_emergency`, and `number_inpatient` display extreme outlier behaviour (max values: 35, 42, and 21 respectively) despite median values near zero.
- `time_in_hospital` has two visible outlier points at 13 and 14 days.

6.4 Skewness Analysis

Quantitative skewness coefficients were computed using `df[numeric_cols].skew()`:

Table 6.1. Feature Skewness Analysis Results

Feature	Skewness Value	Distribution Type
<code>number_emergency</code>	18.33	Heavily Right-Skewed
<code>number_outpatient</code>	9.54	Heavily Right-Skewed
<code>number_inpatient</code>	3.66	Right-Skewed
<code>num_medications</code>	1.49	Right-Skewed
<code>discharge_disposition_id</code>	2.01	Right-Skewed
<code>admission_type_id</code>	1.26	Right-Skewed
<code>num_procedures</code>	1.23	Right-Skewed
<code>patient_nbr</code>	1.02	Right-Skewed
<code>num_lab_procedures</code>	-0.12	Approximately Normal
<code>number_diagnoses</code>	-0.47	Slightly Left-Skewed
<code>encounter_id</code>	-0.15	Approximately Normal

These skewness values have direct implications for future modelling: highly skewed features may require logarithmic or power transformation prior to inclusion in linear or distance-based models.

6.5 Correlation Analysis

A Pearson correlation matrix was computed for all 13 numerical features and visualised as a heatmap with the `coolwarm` colour palette. Notable inter-feature correlations are summarised below:

Table 6.2. Notable Inter-Feature Correlation Coefficients

Feature Pair	Correlation	Strength
<code>num_procedures</code> ↔ <code>num_medications</code>	0.4788	Moderate–Strong
<code>num_medications</code> ↔ <code>time_in_hospital</code>	0.4775	Moderate
<code>admission_type_id</code> ↔ <code>admission_source_id</code>	0.3656	Moderate
<code>time_in_hospital</code> ↔ <code>num_lab_procedures</code>	0.3235	Weak–Moderate
<code>number_emergency</code> ↔ <code>number_inpatient</code>	0.2225	Weak
<code>number_diagnoses</code> ↔ <code>time_in_hospital</code>	0.2660	Weak

The strongest non-trivial correlation (0.4788) between `num_procedures` and `num_medications` is clinically interpretable: patients undergoing more procedures tend to receive more complex medication regimens. The correlation between `num_medications` and `time_in_hospital` (0.4775) indicates that longer hospitalisations are associated with increased pharmacological complexity.

6.6 Feature Interaction — Pair Plot

A comprehensive pair plot was generated using `sns.pairplot()` applied to the full set of 13 numerical features, producing a 13×13 grid of scatter plots and diagonal KDE density curves. The plot confirmed structural observations from prior analyses and served as a comprehensive visual reference for multi-dimensional feature relationships.

7. Key Findings and Insights

7.1 Top 10 Key Findings

The following ten key findings were formally documented in the notebook `05_Top_10_Findings_Part1.ipynb` on 08 June 2026, synthesising observations across all analytical phases:

Table 7.1. Top 10 Key Findings from Diabetes Dataset Analysis

No.	Observation / Finding
1	The dataset contains a sufficient volume of records (101,766 encounters) to support robust statistical analysis and future machine learning applications.
2	Selected healthcare features exhibit missing values, most notably <code>max_glu_serum</code> (94.68% missing) and <code>A1Cresult</code> (83.29% missing), which significantly limits their analytical utility.
3	Duplicate records are entirely absent from the dataset, confirming the integrity of encounter-level data collection.
4	Strong correlational relationships exist between procedure-related variables: <code>num_procedures</code> and <code>num_medications</code> exhibit a correlation of 0.4788, indicating a clinically meaningful association between treatment complexity and medication burden.
5	Several numerical features exhibit pronounced right-skewed distributions. Emergency-related and outpatient visit counts are particularly skewed, suggesting extreme values are present but infrequent.
6	Outliers are present across multiple clinical and administrative indicators. Features such as <code>num_lab_procedures</code> (max: 132), <code>num_medications</code> (max: 81), and <code>time_in_hospital</code> (max: 14) contain extreme values that warrant careful treatment prior to predictive modelling.

No.	Observation / Finding
7	Both categorical and numerical feature types are well-represented, providing analytical flexibility for classification, regression, and clustering approaches.
8	Patient-level characteristics vary significantly across the cohort, reflecting the heterogeneity of diabetic patient populations in real-world healthcare settings.
9	Correlation analysis reveals clinically relevant feature interactions: time in hospital correlates with both medication count and laboratory procedures, suggesting longer admissions involve more complex treatment protocols.
10	The overall dataset structure and feature composition are suitable for the development of future predictive models targeting diabetes-related health outcomes and hospital readmission risk.

7.2 Healthcare Insights Summary

7.2.1 Treatment Complexity Associations

The correlation between `num_procedures` and `num_medications` ($r = 0.479$) suggests that patients undergoing more procedures are systematically placed on more complex medication regimens. This association is clinically coherent and may inform resource allocation models.

7.2.2 Laboratory Investigation Intensity

The strong mean value of `num_lab_procedures` (43.11 per encounter) combined with the near-normal distribution indicates that laboratory investigations are a high-volume standard component of diabetic patient care. The upper tail extending to 132 procedures warrants investigation to identify high-acuity patient subgroups.

7.2.3 Missing Clinical Measurement Data

The near-complete absence of `max_glu_serum` and `A1Cresult` values is a critical finding. Both features represent direct physiological indicators of glycaemic control, yet their systematic absence renders them effectively unusable for predictive modelling without imputation or supplementation.

7.2.4 Readmission Pattern Implications

The multi-class readmitted variable (NO, <30 days, >30 days) provides a structured target for future classification modelling. Observed correlations between hospitalisation duration, inpatient visit counts, and clinical complexity indicators suggest readmission risk may be partially predictable from encounter-level features.

8. Conclusions and Recommendations

This report has documented a thorough and methodical analysis of the Diabetes Dataset conducted over six analytical sessions from 01 to 08 June 2026, progressing through environment setup, dataset understanding, data quality assessment, exploratory analysis, insight generation, and key findings documentation.

8.1 Conclusions

- *The Diabetes Dataset is a rich, large-scale clinical resource containing 101,766 encounter records across 50 features, providing a strong foundation for healthcare data science applications.*
- *The dataset is free of duplicate records and exhibits well-behaved cardinality in most features, indicating a reliable data collection pipeline.*
- *Two critical features — `max_glu_serum` and `A1Cresult` — are substantively incomplete (94.68% and 83.29% missing respectively), severely limiting their analytical utility without imputation.*
- *Moderate to strong correlations between clinical procedure counts, medication complexity, and hospitalisation duration are consistently observed and are clinically interpretable.*
- *Patient visit frequency features exhibit extreme positive skewness, necessitating transformation approaches in subsequent modelling phases.*
- *Outliers are present across multiple features and should be carefully evaluated prior to model training.*

8.2 Recommendations for Future Work

1. *Conduct targeted imputation or exclusion strategies for `max_glu_serum` and `A1Cresult`, documenting the analytical rationale.*
2. *Apply appropriate data transformations (e.g., $\log_{1p}$) to right-skewed features prior to model*

training.

- 3. Encode categorical variables using suitable strategies (label encoding, one-hot encoding, or target encoding) to prepare features for machine learning input.*
- 4. Develop a readmission prediction model using the full feature set, employing readmitted as the multi-class target variable. Baseline models should include logistic regression, decision tree, and random forest classifiers.*
- 5. Investigate medical_specialty and diagnosis code columns (diag_1/2/3) for feature engineering opportunities through ICD code grouping into clinically relevant disease categories.*
- 6. Perform cross-validation and hyperparameter optimisation, evaluating performance using accuracy, F1-score, and AUC-ROC metrics.*
- 7. Document all modelling decisions and results in a final comprehensive project report.*

9. Daily Work Log

The following table provides a structured chronological record of all analytical work completed during the reporting period:

Table 9.1. Chronological Daily Work Log

Date	Notebook	Key Activity	Git Commit
01/06/2026	01_Data_Loading	Project environment setup, Git configuration, dataset acquisition from Kaggle	Created project structure
02/06/2026	01_Data_Loading	Dataset loading, structural exploration, feature summary table generation	Added dataset understanding notebook
03/06/2026	02_Data_Quality	Missing value analysis, heatmap generation, duplicate detection, quality report CSV	Completed data quality assessment
04/06/2026	03_EDA_Visualizat	Histograms, KDE plots, box-plots, correlation heatmap, pair plots, skewness analysis	Completed EDA and visualizations
05/06/2026	04_Insights_Repor	Insight generation, correlation review, structured observations, summary documentation	Completed Week 1 analysis and reporting
08/06/2026	05_Top_10_Finding	Top 10 key findings identification, findings DataFrame creation, final documentation	Added top 10 key findings analysis

All notebooks are available within the `notebooks/` directory of the GitHub repository at <https://github.com/praneeth-tech/HF-PMS.git>.

10. References and Resources

10.1 Primary Dataset

Brandao, J. (2014). *Diabetes 130-US Hospitals for Years 1999–2008 Dataset*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/brandao/diabetes>

10.2 Libraries and Frameworks

McKinney, W. (2010). *Data structures for statistical computing in Python*. *Proceedings of the 9th Python in Science Conference*, 51–56. [Pandas]

Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment*. *Computing in Science & Engineering*, 9(3), 90–95.

Waskom, M. L. (2021). *seaborn: Statistical data visualization*. *Journal of Open Source Software*, 6(60), 3021.

Harris, C. R., et al. (2020). *Array programming with NumPy*. *Nature*, 585, 357–362.

10.3 Version Control and Repository

GitHub Repository (HF-PMS): <https://github.com/praneeth-tech/HF-PMS.git>

All analytical notebooks, visualisation outputs, and documentation are archived within the above repository and are accessible for academic review.

10.4 Development Environment

Google Colaboratory (Colab): <https://colab.research.google.com> — cloud-based Jupyter notebook environment used for all analytical computations.

Overleaf: <https://overleaf.com> — LaTeX-based documentation platform used for formal

report preparation.

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